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Physiological characteristics predictive of passing military physical employment standard tasks for ground close combat occupations in men and women

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Abstract

Challenges for some women meeting the physical employment standards (PES) for ground close combat (GCC) roles stem from physical fitness and anthropometric characteristics. The purpose of this study was to identify the modifiable and non-modifiable characteristics predictive of passing GCC-based PES tasks and determine the modifiable characteristics suitable to overcome nonmodifiable limitations. 107 adults (46 women) underwent multiday testing assessing regional and total lean mass (LM), percent body fat (BF%), aerobic capacity ($\dot{V}O_{2peak}$), strength, power, and PES performance. Predictors with p-value <0.200 were included in stepwise logistic regression analysis or binary logistic regression when outcomes among sexes were insufficient. Relative and absolute arm LM (OR: 4.617–8.522, $p < 0.05$), leg LM (OR: 2.463, $p < 0.05$), and upper body power (OR: 2.061, $p < 0.05$) predicted medicine ball chest throw success. Relative and absolute arm LM (OR: 3.734–11.694, $p < 0.05$), absolute trunk LM (OR: 2.576, $p < 0.05$), and leg LM (OR: 2.088, $p < 0.05$) predicted casualty drag success. Upper body power (OR: 3.910, $p < 0.05$), absolute trunk LM (OR: 2.387, $p < 0.05$), leg LM (OR: 2.290, $p < 0.05$), and total LM (OR: 1.830, $p < 0.05$) predicted maximum single lift success. Relative and absolute arm LM (OR: 3.488–7.377, $p < 0.05$), leg LM (OR: 1.965, $p < 0.05$), and upper body power (OR: 1.957, $p < 0.05$) predicted water can carry success. %BF (OR: 0.814, $p = 0.007$), $\dot{V}O_{2peak}$ (OR: 1.160, $p = 0.031$), and lower body strength (OR: 1.059, $p < 0.001$) predicted repeated lift and carry success. $\dot{V}O_{2peak}$ (OR: 1.540, $p < 0.001$) predicted 2-km ruck march success. Modifiable characteristics were the strongest predictors for GCC-based PES task success to warrant their improvement for enhancing PES performance for women.

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KEYWORDS
body composition, fitness, gender, prediction, strength

Highlights

- Women in GCC roles require sufficient military-specific physical fitness and anthropometric parameters, independent of sex, for successfully passing GCC-based PES tasks as part of assessment testing and selection procedures for GCC roles.
- Passing GCC-based PES tasks involving heavy lifting, dragging, pushing/throwing, and carrying may benefit from increased upper body (arm and trunk) and lower body lean mass and upper body power and maximal strength, whereas passing material handling tasks may benefit from lower percent body fat, greater lower body muscular strength, and higher aerobic capacity with the latter important for passing load carriage tasks.
- Military trainers should target modifiable characteristics, including upper body lean mass, power, and maximal strength to improve GCC-based PES task performance in women.

1 | INTRODUCTION

The recent uplift of bans that prohibited the entrance of women in ground close combat (GCC) roles—roles which involve engaging with an enemy on the ground using weaponry and encountering physical contact with hostile force personnel (Fitriani et al., 2016)—in several countries, such as the United States, United Kingdom, and Australia, has led to a significant increase of women failing to meet the minimum military-specific physical fitness requirements as part of the assessment testing and selection procedures for GCC roles in Australia (Drain et al., 2015) and the United States (Beynon, 2021). Failure of women meeting the minimum physical fitness requirements for such roles has been suggested to be attributed to physical fitness and anthropometric disadvantages compared to men (Conkright et al., 2022a; Nindl, 2015; Sterczala et al., 2023). Further investigation of effective strategies to overcome these disadvantages and enhance the success rate of women meeting physical fitness requirements for GCC roles is warranted.

All eligible service members, regardless of sex, must perform a standardized, gender-neutral, and job-specific physical employment standard (PES) assessment as part of assessment testing and selection procedures for GCC roles in the United States (Beynon, 2021), United Kingdom (Sterczala et al., 2023), and Australia (Drain et al., 2015). These assessments aim to predict GCC-specific physical and tactical readiness through the performance of simulated GCC-based tasks utilizing several physical fitness domains essential for successful military occupational task performance. Such domains include aerobic capacity, upper and lower body muscular strength, endurance, and power, which, when taken together, aim to reflect the minimum role-related physical fitness for GCC roles (Knapik, 1997; Vaara et al., 2022). GCC roles involve the frequent act of performing military occupational tasks, such as load carriage, manual material handling, equipment carrying and dragging heavy loads, obstacle traversal, marching, running, and sprinting over various distances (Vaara et al., 2022). Hence, these roles are recognized as physically demanding (Friedl et al., 2023), and PES assessments are designed to measure the physical requirements necessary to meet the minimum physical demands of a job-role (Brown et al., 2019). Assessment of physical requirements is most effectively accomplished through the measurement of binary PES outcomes that categorize individuals as either passing or failing the PES task based on predetermined and task-specific standards (Petersen et al., 2016). However, despite successful efforts in evaluating PES task performance in women for GCC roles (Fitriani et al., 2016), the rates of women passing GCC-based PES tasks remain a cause for concern (Beynon, 2021; Drain et al., 2015; Hall et al., 2022).

For instance, after the introduction of the Army Combat Fitness Test (ACFT) as the new United States Army physical fitness standard in 2020, approximately 44% of female soldiers compared to 7% of male soldiers failed to meet the minimum GCC-based PES over a 6-month period with the greatest failure rates of female soldiers attributed to the standing power throw (15%), two-mile run (22%), and leg tuck (22%) as a result of lower absolute upper body power, aerobic capacity, and upper body strength, respectively (Beynon, 2021). Additionally, only 20% (n = 4) of a group of female Australian Army recruits completed the minimum GCC-based PES of a one-repetition maximum (1RM) box lift onto a 1.50 m-high platform, whereas 99.4% (n = 153) of the men were able to achieve the standard due to greater absolute strength and taller stature (Drain et al., 2015). Addressing the deficiency of GCC-based PES performance in women is crucial for enhancing military occupational task performance in GCC roles. Understanding the modifiable characteristics (e.g., physical fitness and body composition) and nonmodifiable characteristics (e.g., demographics and anthropometric traits) conducive to successful GCC-based PES performance may be critical to improve GCC PES success rates in women.

The physiologic differences in modifiable and nonmodifiable characteristics between men and women in military services are well-established (Greeves, 2015). In general, women exhibit lower physical fitness (Kraemer et al., 2001), including upper and lower body strength (Epstein et al., 2013; Nindl et al., 2017), expressed in both absolute (Lockie et al., 2020) and relative terms (Comfort et al., 2024; Nimphius et al., 2019), muscular endurance and power (Epstein et al., 2013), aerobic and anaerobic endurance (Hill et al., 2014; Pate et al., 1984), as

well as lower lean mass, higher fat mass, smaller skeletons, and shorter height and limb lengths compared to men, which can hinder military occupational task performance (Conkright et al., 2022a; Nindl et al., 2017) and increase the risk of noncombat-related musculoskeletal injury (Greeves, 2015). However, recent studies demonstrate an overlap in physical performance related to PES between sexes attributed to enhanced physical fitness characteristics, where the highest-performing women achieve greater physical fitness performance scores than the lowest-performing men (Reilly et al., 2019). Furthermore, previous studies demonstrate that enhanced modifiable characteristics, such as physical fitness and body composition parameters, including lean mass, fat mass, and percent body fat, can improve military-specific task performance in women (Knapik, 1997; Reilly et al., 2016) and minimize the physical performance gap between sexes (Kraemer et al., 2001; Nindl et al., 2017). However, despite previous research on the predictive value of modifiable and nonmodifiable characteristics on military-specific PES performance utilizing continuous scales (Agostinelli et al., 2022; Reilly et al., 2016) and mean differences between performers (Tingelstad et al., 2016), it remains unknown which specific characteristics predict the discrete outcome of passing or failing a GCC-based PES assessment. Further, the assessment of modifiable characteristics necessary for women to overcome nonmodifiable limitations, such as age, height, and sex, has yet to be investigated (Nindl, 2015). Understanding the modifiable and nonmodifiable characteristics conducive to passing a GCC-based PES assessment may aid in the physical preparation of incoming female military recruits and increase the appreciation of targeting pertinent modifiable characteristics to overcome nonmodifiable delimiters observed in women for improving GCC PES performance.

Hence, the aim of this study was to identify the modifiable and nonmodifiable characteristics conducive to passing a GCC-based PES assessment and determine the modifiable characteristics suitable for overcoming nonmodifiable delimiters including age, height, and sex. Drawing from previous research (Agostinelli et al., 2022; Chassé et al., 2019; Nindl, 2015; Tingelstad et al., 2016), it was hypothesized that sufficient upper body (arm and trunk) lean mass and upper body muscular power and maximal strength would be necessary to pass GCC-based tasks involving heavy lifting, dragging, pushing and throwing, and carrying heavy loads (H1). Additionally, it was hypothesized that passing material handling and load carriage tasks would require higher aerobic capacity, lower body fat percentage, and greater upper and lower body muscular strength (H2). Finally, it was hypothesized that modifiable characteristics would have stronger predictive value for PES outcomes across all tasks compared to nonmodifiable characteristics (H3).

2 | MATERIALS AND METHODS

2.1 | Experimental design

Participants completed a laboratory-based multiday testing battery distributed over four nonconsecutive days (referred to as Day A-D) occurring within a timeframe of 10 to 14 days. The testing battery was a component of the Soldier Performance and Readiness as Tactical Athletes (SPARTA) parent study designed to investigate the effectiveness of a prospective periodized strength and power resistance training program coupled with concurrent interval training on musculoskeletal and performance adaptations within and between sexes as measured by a PES evaluation specific to GCC roles for both men and women (Ahamed et al., 2021; Conkright et al., 2022b; Sterczala et al., 2023). Consequently, the objective of the current study was to conduct a detailed cross-sectional analysis of the modifiable characteristics (e.g., physical fitness and body composition) and nonmodifiable characteristics (e.g., demographics and anthropometric traits) in the context of GCC-based PES performance from data collected during the laboratory-based multiday testing regimen.

Modifiable characteristics, including body composition and traditional physical fitness parameters, such as upper and lower body power and maximal strength and aerobic capacity, and nonmodifiable characteristics were evaluated on Day A, Day B, and Day D. Military occupational performance was assessed on Day C by the performance of a GCC-based PES assessment comprising a set of GCC-based PES tasks based on protocols from those recently developed by the British Army for GCC roles now implemented as the Role Fitness Tests for Soldiers (RFT(S)), Role Fitness Tests for Entry (RFT(E)), and Role Fitness Tests for Basic Training (RFT(BT)). The PES assessment comprised a set of GCC-based PES tasks designed to represent the diverse physical demands of military service through the performance of whole-body and common military exercises relevant to GCC roles (New Physical Employment Standards, 2020). The PES tasks included a seated medicine ball chest throw, simulated casualty drag, single sandbag lift to a raised platform, water can carry, repeated lift and carry, and a 2-km ruck march. All tasks comprised predetermined gender-neutral and age-independent standards to reflect realistic GCC demands. Approximately 24–48 h separated Days A and B and Days C and D, while at least 72 h separated Days B and C. Prior to each test day, a dynamic warmup was performed to prepare participants for performance testing. To control the fatiguing effects of exercise, within-session tests were performed in the order of least fatiguing to most fatiguing to limit confounding factors.

2.2 | Participants

One hundred and seven men and women with available data from the laboratory-based multiday testing battery as part of the SPARTA parent study were included in the current study (Sterczala et al., 2023). Participants were recruited within a large metropolitan area by flyer, email, and word-of-mouth recruitment methods and met eligibility criteria for study enrollment. To be eligible, participants had to meet the following criteria: (1) be aged between 18 and 35, (2) have no musculoskeletal injuries or conditions that could affect their physical and/or PES performance, and (3) engage in

moderate to high-intensity physical activity at least three times per week. Exclusion criteria included: (1) diagnosis of a medical condition, whether physical or psychological, that hindered their ability to participate in exercise, (2) sustained a musculoskeletal injury in the past 2 years that prevented physical activity for at least 1 month, (3) current use of medications, such as anticoagulants or those affecting hormone levels (excluding birth control), (4) body mass fluctuation greater than 4.5 kg in the preceding 2 months, (5) current pregnancy or pregnancy during the study, (6) history of heart conditions or high blood pressure, (7) experiencing chest pain during rest, daily activities, or physical activity, and (8) history of drug addiction or regular use of recreational drugs. Participants were recreationally active, civilian students and young professionals without prior military training experience or completion. Pre-intervention training history questionnaires indicated that participants exercised 4.4 ± 1.1 times per week for a total of 5.8 ± 3.6 h per week with endurance and resistance-training specific physical activity by sex as reported previously (Sterczala et al., 2023). Prior to enrollment, all volunteers were informed of the risks and provided verbal and written informed consent. The investigation was approved by the University of Pittsburgh Institutional Review Board and United Kingdom Ministry of Defense Research Ethics Committee (903/MODREC/18).

2.3 | Test Day A

2.3.1 | Body composition

Participants fasted for at least 8 h and refrained from physical activity for at least 24 h before body composition assessment. A calibrated scale (Healthometer Professional Scale 349KLX, Boca Raton, FL, USA) measured participant body mass (kg), while a stadiometer (Seca 216, Hamburg, GER) measured height (cm). Dual-energy X-ray absorptiometry (DXA) scans (Lunar iDXA, GE Healthcare, Madison, WI, USA) assessed regional and total percent body fat (%BF), absolute lean mass (LM; kg), and absolute fat mass (FM; kg). Regional LM to total LM ratios were calculated by dividing the regional LM for each segment (e.g., arm, trunk, and leg) by the total LM and multiplied by 100 to express as a percentage ([regional LM/total LM] × 100), which enabled an operationalization for relative LM values (%). Scans were repeated if any artifacts or incongruity were detected. A trained operator performed daily calibration using the GE Lunar calibration phantom. All images were obtained and analyzed using EnCore software version 15 (GE Healthcare, Madison WI, USA). The in vivo coefficient of variation for soft tissue and %BF measured by this DXA model ranges from 0.70% to 2.07%. Although DXA models enable the identification of android and gynoid fat distribution and visceral adipose tissue such markers of central adiposity were not measured as modifiable characteristics of interest due to increased evidence of their regulation from genetic traits that vary across ethnic groups (Sun et al., 2021). Hence, the inclusion of the aforementioned markers of subcutaneous adipose tissue stemmed from their susceptibility to be influenced by modifiable factors (e.g., aerobic or resistance exercise) irrespective of ethnicity or genetics as indicated by large-scale epidemiological studies (Ramírez-Vélez et al., 2020).

2.3.2 | Upper body power

Isokinetic dynamometer (Universal Pro T-Base, Biodex, Shirley, NY, USA) assessed upper body power using maximal isokinetic exercise at a speed of 120° s^{-1}. Participants positioned themselves in the supine seat with elbows at 90° and performed one set of five maximal repetitions per limb of pushing and pulling the handle away and toward the chest through full range of motion. Average power was calculated by the summed work per limb directed away from the body divided by total duration in seconds and averaged across limbs ($W = [N{\cdot}m{\cdot}s^{-1}_{Limb1} + N{\cdot}m{\cdot}s^{-1}_{Limb2}]/2$). Calculated average power was normalized to body mass to operationalize relative values ($W{\cdot}kg^{-1}$).

2.3.3 | Aerobic capacity

Participants performed a standard incremental Bruce protocol (Bruce, 1971) on a calibrated treadmill (Woodway, Waukesha, WI, USA) with a metabolic system (Parvo TrueOne, Salt Lake City, UT, USA) to volitional failure. Aerobic capacity was assessed as peak oxygen consumption ($\dot{V}O_{2peak}$) relative to body mass measured by indirect calorimetry (Parvo TrueOne, Salt Lake City, UT, USA) during the protocol to operationalize relative values ($mL{\cdot}kg^{-1}{\cdot}min^{-1}$).

2.4 | Test Day B

2.4.1 | Maximal strength

Upper and lower body maximal strength was assessed using one-repetition maximum (1RM) for the back squat, bench press, and conventional deadlift. Testing followed established protocols by the National Strength and Conditioning Association and adhered to International Powerlifting Federation guidelines (IPF Technical Rules Book 2024 24 Jan pdf, 2024). Loads were increased and decreased accordingly until the participant successfully completed a 1RM (kg). Between all warmup sets and 1RM attempts, two-min rest periods were provided.

2.5 | Test Day C

2.5.1 | Physical employment standard assessment

The PES assessment involved the performance of simulated GCC-specific tasks based on protocols from those recently developed by the British Army for GCC roles now implemented as the RFT(S), RFT(E), and RFT(BT) in the following order: seated medicine ball chest throw (MBCT), simulated casualty drag (CD), maximum single lift (MSL),

water can carry (WCC), repeated lift and carry (RLC), and 2-km ruck march (2KRM). All tasks took place in a temperature-controlled, pressurized air structure on a crumb rubber-filled turf field. Participants were familiarized with the PES tasks at the pretraining Day B following 1RM testing. Prior to test Day C, participants were instructed to abstain from physical activity for a minimum of 48 h. Except for the MBCT, participants wore body armor (11 kg) and a tactical helmet during PES testing to mimic GCC soldier conditions.

2.5.2 | Seated medicine ball chest throw

Participants sat with their back and hips against a wall while holding a 4 kg medicine ball against their chest. On command, they pushed the ball upward and outward to achieve a maximum throwing distance (m) measured from the base of the wall to where the ball landed. Throws were disqualified if participants moved their back off the wall, lifted their legs, or deviated from a straight path. Maximum distance from three trial attempts determined pass (≥3.1 m) or fail (<3.1 m).

2.5.3 | Casualty drag

Participants dragged a 111 kg casualty dummy backward over 20 m as quickly as possible. Participants were allowed to stop and readjust their grip during the test if needed. Time to completion was graded as pass (≤60 s) or fail (>60 s).

2.5.4 | Maximum single lift

Participants lifted power (sand) bags from the floor to a 1.49 m-high platform. Power bag mass started at 20 kg and increased 5 kg with each successful lift. Lifts continued with one-minute rest periods in between until failure or when the participant reached a 60 kg maximal load. Participants were allowed to use their knee to assist in lifting the power bag. No repeat attempts were granted for a failed lift. The maximal successful lift mass was recorded and graded as pass (≥20 kg) or fail (<20 kg).

2.5.5 | Water can carry

Participants carried two 22 kg sand-filled water cans 240 m (4 × 60 m laps). Participants could place the water cans down as needed to complete the task. Time to completion was recorded and graded as pass (≤4 min) or fail (>4 min).

2.5.6 | Repeated lift and carry

After a five-min rest interval, participants picked up a 20 kg sandbag from the floor, carried it around a cone 15 m away, returned to the start, placed the sandbag on a 1.49 m-high platform, returned the sandbag to the floor, and completed a 30-m run without the sandbag around the cone. This sequence was considered one repetition, and a total of 20 repetitions were performed. Test time started when the participant touched the bag and ended after completing all repetitions. Participants were not allowed to throw the bag on the platform or drop it uncontrollably on the floor. Time to complete the task was graded as pass (≤14 min) or fail (>14 min).

2.5.7 | 2-km ruck March

Following a 10-min rest interval, participants completed 10 laps around a 200-m course as fast as possible for a total of 2 km while carrying 25 kg of equipment (body armor, tactical helmet, slung weapon, and loaded rucksack). Time to completion was graded as pass (≤15 min) or fail (>15 min).

2.6 | Test Day D

2.6.1 | Lower body peak power

Lower body peak power was assessed by a maximal bilateral squat jump on dual force plates (Kistler Type 9286B, Kistler Inc., Winterthur, SWTZ) sampling at 1000 Hz implemented with Vicon software (Vicon Nexus, Yarnton, UK) for data collection. Participants performed the maximal bilateral squat jump under a standard barbell instrumented in a smith machine (Star Trac Max Rack, Star Trac Enterprise LLC, Southfield, MI, USA) weighted 30% of the 1RM achieved during the back squat as the accepted mass to elicit peak power outputs during fixed-form (e.g., smith machine) squat jumps in both men and women (Thomas et al., 2007). Participants positioned one foot on each force plate under the weighted bar, squatted down to touch a box adjusted by height to enable a 90˚ bend at the knee joint, paused on the box for at least two seconds to enable a level waveform, and executed a bilateral vertical jump with maximum speed to obtain maximal vertical height. Peak power was calculated as the maximum power output achieved in the force-velocity profile during the concentric portion of the waveform. The concentric portion of the waveform was considered from the first instance the normalized force (force minus system mass in N) crossed 0 (upward slope) and then crossed 0 again (downward slope). Peak power output of highest value from three trials expressed relative to body mass ($W \cdot kg^{-1}$) was used for analysis due to its association with the rate of force development in the lower body independent of confounders including body size, height, and mass between sexes (Turner et al., 2015). Participants repeated the task if they landed outside the confines of the force plates. MATLAB (The Mathworks, Inc., Watertown, MA, USA) was used for data reduction and analysis using custom scripts.

2.7 | Statistical analyses

Statistical analyses were performed in SPSS version 28.0 (IBM Corp, Armonk, NY, USA) and Stata/SE 17.0 for Windows (StataCorp LLC, College Statoin, TX, USA). Descriptive statistics were presented as mean ± standard deviation, median (Q1 and Q3), or percentage as appropriate. Independent samples *t*-tests or Mann–Whitney U tests compared body composition, fitness, and performance variables between sexes. Nonmodifiable (age, height, and sex) and modifiable (body mass, regional and total LM, regional and total FM, regional to total LM ratios, %BF, $\dot{V}O_{2peak}$, upper body power and lower body peak power, 1RM back squat, 1RM bench press, and 1RM deadlift) predictors for passing PES tasks were determined a priori via job task analysis by listing the essential characteristics thought conducive to passing each task, which were subject to reassignment or removal after discussion (Brown et al., 2019). Predictors with a *p*-value <0.200 in the simple logistic regression analysis were considered for inclusion in a backward stepwise multiple logistic regression analysis. In case of PES outcomes that comprised not enough outcomes among either sex or disallowed executing multiple logistic regression, simple binary logistic regression analysis was performed. Statistical significance was set a priori $\alpha = 0.05$, two-tailed test.

3 | RESULTS

3.1 | Participants

Results of physical fitness, body composition, and PES performances by sex are shown in Table 1. Men were taller and heavier and had significantly greater arm LM, trunk LM, leg LM, arm to total LM ratio, trunk LM to total LM ratio, and total LM than the women. Additionally, men showed significantly lower %BF and leg FM and outperformed the women in all fitness and PES tasks (Table 1). Men passed PES tasks to a greater proportion than the women (Table 2).

3.2 | Modifiable and nonmodifiable characteristics predictive of passing PES tasks

3.2.1 | Repeated lift and carry

Results of the backward stepwise multiple logistic regression analysis of characteristics predicting passing the RLC task are shown in Table 3. Inclusion of sex, height, body mass, total LM, arm LM, trunk LM, leg LM, total FM, %BF, $\dot{V}O_{2peak}$, and 1RM deadlift and 1RM back squat revealed total FM, body mass, 1RM back squat, and arm LM significantly predicted task success (Model 1). However, multicollinearity between total FM, body mass, and arm LM (VIFs >10) led to a second

TABLE 1 Physical fitness, body composition, and military occupational task performances by sex.

Characteristic	Men (*n* = 61)	Women (*n* = 46)
Nonmodifiable		
Age (y)	27.20 ± 5.11	26.83 ± 4.68
Height (cm)**	**178.92 ± 7.36**	**165.62 ± 5.91**
Modifiable		
Body mass (kg)**	**84.65 ± 15.75**	**66.05 ± 10.57**
%BF (%)**	**24.42 ± 7.75**	**32.08 ± 6.73**
Arm LM (kg)**	**7.98 ± 1.47**	**4.49 ± 0.70**
Trunk LM (kg)**	**27.07 ± 3.30**	**19.56 ± 2.38**
Leg LM (kg)**	**21.57 ± 3.43**	**14.92 ± 2.26**
Total LM (kg)**	**60.17 ± 7.67**	**41.72 ± 4.87**
Arm LM to total LM ratio (%)**	**13.20 ± 1.12**	**10.75 ± 0.94**
Trunk LM to total LM ratio (%)**	**45.06 ± 2.24**	**46.66 ± 2.22**
Leg LM to total LM ratio (%)	35.76 ± 2.13	35.46 ± 2.39
Arm FM (kg)	2.16 ± 0.98	2.42 ± 0.79
Trunk FM (kg)	11.59 ± 6.32	10.01 ± 4.47
Leg FM (kg)**	**6.38 ± 2.59**	**7.66 ± 2.61**
Total FM (kg)	21.44 ± 10.21	21.68 ± 7.43
$\dot{V}O_{2peak}$ ($mL \cdot kg^{-1} \cdot min^{-1}$)**	**49.10 ± 7.94**	**40.75 ± 6.58**
Upper body power ($W \cdot kg^{-1}$)**	**5.36 ± 1.98**	**3.60 ± 1.26**
1 RM back squat (kg)**	**104.01 ± 32.29**	**56.16 ± 17.00**
1 RM bench press (kg)**	**81.22 ± 22.95**	**35.46 ± 7.72**
1 RM deadlift (kg)**	**121.88 ± 31.41**	**73.94 ± 14.62**
Lower body peak power ($W \cdot kg^{-1}$)**	**38.16 ± 8.01**	**32.46 ± 8.91**
Task performance		
MBCT (m)**	**4.42 ± 0.70**	**2.78 ± 0.34**
CD (s)**	**21.26 ± 5.58**	**43.39 ± 9.90**
MSL (kg)**	**48.36 ± 9.43**	**25.25 ± 4.52**
WCC (min)**	**2.65 ± 0.95**	**5.45 ± 1.65**
RLC (min)**	**13.00 ± 2.11**	**16.66 ± 2.81**
2KRM (min)*	**10.77 ± 2.43**	**12.52 ± 2.77**

Note: Values are expressed as mean ± SD. Independent samples *t*-test was performed to determine statistically significant differences between sex denoted in boldface with asterisk where (*) $p < 0.05$ and (**) $p < 0.001$.

Abbreviations: %BF, percent body fat; 1RM, one-repetition maximum; 2KRM, 2 km ruck march task; CD, casualty drag task; FM, fat mass; LM, lean mass; MBCT, medicine ball chest throw task; MSL, maximum single lift task; RLC, repeated lift and carry task; $\dot{V}O2peak$, aerobic capacity; WCC, water can carry task.

TABLE 2 Proportion of participants who passed or failed or missed PES tasks by sex.

Task	Men pass	Women pass	Men fail	Women fail	Men missing	Women missing	Total pass	Total fail	Total missing
MBCT	61 (100.0%)	15 (32.6%)	0 (0.0%)	31 (67.4%)	0 (0.0%)	0 (0.0%)	76 (71.0%)	31 (29.0%)	0 (0.0%)
CD	60 (98.4%)	14 (30.4%)	1 (1.6%)	32 (69.6%)	0 (0.0%)	0 (0.0%)	74 (69.2%)	33 (30.8%)	0 (0.0%)
MSL	61 (100.0%)	40 (87.0%)	0 (0.0%)	6 (13.0%)	0 (0.0%)	0 (0.0%)	101 (94.4%)	6 (5.6%)	0 (0.0%)
WCC	60 (98.4%)	41 (89.1%)	1 (1.6%)	5 (10.9%)	0 (0.0%)	0 (0.0%)	101 (94.4%)	6 (5.6%)	0 (0.0%)
RLC	43 (58.1%)	6 (13.0%)	18 (41.9%)	40 (87.0%)	0 (0.0%)	0 (0.0%)	49 (45.8%)	58 (54.2%)	0 (0.0%)
2KRM	55 (90.2%)	37 (80.4%)	6 (9.8%)	8 (17.4%)	0 (0.0%)	1 (2.2%)	92 (86.8%)	14 (12.3%)	1 (0.9%)

Abbreviations: 2KRM, 2 km ruck march task; CD, casualty drag task; MBCT, medicine ball chest throw task; MSL, maximum single lift task; RLC, repeated lift and carry task; WCC, water can carry task.

TABLE 3 Results of multiple binary logistic regression models predicting passing repeated lift and carry task.

Model	Predictor variable	β	Odds ratio	Std. Error	Z	*p*-value	95% CI
Model 1	Total FM(kg)	−1.050	0.350	0.095	−3.85	**<0.001**	0.21, 0.60
	Body mass (kg)	0.555	1.742	0.282	3.42	**0.001**	1.27, 2.39
	1RM back squat (kg)	0.112	1.118	0.040	3.13	**0.002**	1.04, 1.20
	Arm LM (kg)	−2.442	0.087	0.074	−2.87	**0.004**	0.02, 0.46
	Constant	−14.415	5.49×10^{-7}	2.10×10^{-6}	−3.76	**<0.001**	3.02×10^{-10}, 0.09
Model 2	$\dot{V}O_{2peak}$ (mL·kg^{-1}·min^{-1})	0.148	1.160	0.080	2.15	**0.031**	1.01, 1.33
	1RM back squat (kg)	0.057	1.059	0.014	4.24	**<0.001**	1.03, 1.09
	%BF (%)	−0.206	0.814	0.062	−2.70	**0.007**	0.70, 0.94
	Constant	−6.215	0.002	0.007	−1.41	0.160	1.78×10^{-7}, 12.91

Note: Predictors included in Model 1 involved height (cm), body mass (kg), sex, total LM (kg), arm LM (kg), leg LM (kg), trunk LM (kg), total FM (kg), %BF (%), $\dot{V}O2_{peak}$ (mL kg^{-1} min^{-1}), 1RM deadlift (kg), and 1RM back squat (kg). Predictors included in Model 2 involved height (cm), body mass (kg), sex, total LM (kg), %BF (%), $\dot{V}O_{2peak}$ (mL kg^{-1} min^{-1}), 1 RM deadlift (kg), and 1RM back squat (kg). Constant estimates baseline odds. Bolded values indicate statistical significance ($p < 0.05$).

Abbreviations: %BF, percent body fat.; 1RM, one-repetition maximum; FM, fat mass; LM, lean mass; $\dot{V}O2peak$, aerobic capacity.

model, including sex, height, body mass, total LM, %BF, $\dot{V}O_{2peak}$, 1RM deadlift, and 1RM back squat, resulting in %BF, $\dot{V}O_{2peak}$, and 1RM back squat significantly predicting task success (Model 2).

3.2.2 | 2 km ruck March

Results of the backward stepwise multiple logistic regression analysis of characteristics predicting passing the 2KRM task are shown in Table 4. Inclusion of total FM, %BF, $\dot{V}O_{2peak}$, and 1RM back squat revealed total FM, %BF, $\dot{V}O_{2peak}$, and 1RM back squat significantly predicted task success (Model 1). However, multicollinearity between total FM and %BF (VIFs 6.79 and 8.20, respectively) led to a second model, including %BF, $\dot{V}O_{2peak}$, and 1RM back squat, resulting in $\dot{V}O_{2peak}$ significantly predicting task success (Model 2).

3.2.3 | Water can carry

Results of simple binary logistic regression analysis of characteristics predicting passing the remaining tasks are shown in Table 5. Significant characteristics predictive of passing WCC, from strongest to weakest, included arm LM (OR: 7.377 [1.41, 38.55]), arm to total LM ratio (OR: 3.488 [1.36, 8.94]), leg LM (OR: 1.965 [1.16, 3.32]), upper body power (OR: 1.957 [1.01, 3.78]), trunk LM (OR: 1.929 [1.10, 3.37]), total LM (OR: 1.432 [1.05, 1.95]), age (OR: 1.270 [1.01, 1.61]), height (OR: 1.178 [1.04, 1.34]), body mass (OR: 1.134 [1.02, 1.26]), 1RM bench press (OR: 1.086 [1.01, 1.17]), and 1RM deadlift (OR: 1.049 [1.00, 1.10]) (Table 5).

3.2.4 | Seated medicine ball chest throw

Significant characteristics predictive of passing the MBCT, from strongest to weakest, included arm LM (OR: 8.522 [3.28, 22.15]), arm to total LM ratio (OR: 4.617 [2.53, 8.42]), leg LM (OR: 2.463 [1.71, 3.55]), upper body power (OR: 2.061 [1.46, 2.91]), trunk LM (OR: 1.782 [1.42, 2.24]), total LM (OR: 1.347 [1.19, 1.52]), trunk to total LM ratio (OR: 0.675 [0.54, 0.84]), leg to total LM ratio (OR: 1.310 [1.07, 1.61]), 1RM bench press (OR: 1.254 [1.12, 1.41]), height (OR: 1.211 [1.12, 1.31]), $\dot{V}O_{2peak}$ (OR: 1.132 [1.06, 1.21]), body mass (OR: 1.131 [1.07, 1.19]), lower body peak power (OR: 1.106 [1.03, 1.19]),

TABLE 4 Results of multiple binary logistic regression models predicting passing 2-km ruck march task.

Model	Predictor variable	β	Odds ratio	Std. Error	Z	*p*-value	95% CI
Model 1	Total FM (kg)	−0.395	0.674	0.115	−2.32	**0.020**	0.48, 0.94
	%BF (%)	0.434	1.574	0.363	1.97	**0.049**	1.01, 2.47
	$\dot{V}O_{2peak}$ (mL·kg^{-1}·min^{-1})	0.512	1.668	0.294	2.91	**0.004**	1.18, 2.36
	1RM back squat (kg)	0.059	1.061	0.031	1.97	**0.048**	1.00, 1.12
	Constant	−26.798	2.30×10^{-12}	2.50×10^{-11}	−2.47	**0.013**	1.35×10^{-21}, 0.04
Model 2	$\dot{V}O_{2peak}$ (mL·kg^{-1}·min^{-1})	0.432	1.540	0.179	3.71	**<0.001**	1.23, 1.93
	Constant	−15.060	2.88×10^{-7}	1.22×10^{-6}	−3.55	**<0.001**	7.12×10^{-11}, 0.01

Note: Predictors of Model 1 included total fat mass (kg), percent body fat (%), $\dot{V}O_{2peak}$ (mL kg^{-1} min^{-1}), and back squat (kg). Predictors for Model 2 included percent body fat (%), $\dot{V}O_{2peak}$ (mL kg^{-1} min^{-1}), and back squat (kg). Constant estimates baseline odds. Bolded values indicate statistical significance ($p < 0.05$).

Abbreviations: %BF, percent body fat.; 1RM, one-repetition maximum; FM, fat mass; LM, lean mass; V̇O2peak, aerobic capacity.

TABLE 5 Results of simple binary logistic regression analysis of modifiable and nonmodifiable characteristics predicting PES task success.

Characteristics	Casualty drag Odds ratio (95% CI)	Maximum single lift Odds ratio (95% CI)	Water can carry Odds ratio (95% CI)	Medicine ball chest throw Odds ratio (95% CI)
Modifiable				
Body mass (kg)	**1.142 (1.08, 1.21)**	**1.146 (1.03, 1.28)**	**1.134 (1.02, 1.26)**	**1.131 (1.07, 1.19)**
Total LM (kg)	**1.515 (1.27, 1.81)**	**1.830 (1.15, 2.92)**	**1.432 (1.05, 1.95)**	**1.347 (1.19, 1.52)**
Total FM (kg)	1.009 (0.96, 1.06)	1.011 (0.92, 1.11)	1.018 (0.92, 1.12)	1.009 (0.96, 1.06)
Arm LM (kg)	**11.694 (3.84, 35.60)**	1.014 (0.99, 1.03)	**7.377 (1.41, 38.55)**	**8.522 (3.28, 22.15)**
Trunk LM (kg)	**2.576 (1.70, 3.91)**	**2.387 (1.17, 4.88)**	**1.929 (1.10, 3.37)**	**1.782 (1.42, 2.24)**
Leg LM (kg)	**2.088 (1.57, 2.78)**	**2.290 (1.22, 4.31)**	**1.965 (1.16, 3.32)**	**2.463 (1.71, 3.55)**
Arm FM (kg)	0.899 (0.57, 1.41)	0.787 (0.33, 1.86)	0.874 (0.36, 2.11)	0.842 (0.53, 1.33)
Trunk FM (kg)	1.062 (0.98, 1.15)	1.065 (0.90, 1.27)	1.061 (0.89, 1.26)	1.059 (0.98, 1.15)
Leg FM (kg)	**0.837 (0.72, 0.98)**	0.896 (0.68, 1.18)	0.946 (0.71, 1.26)	0.860 (0.74, 1.01)
Arm LM to total LM ratio (%)	**3.734 (2.24, 6.23)**	**1.040 (1.01, 1.07)**	**3.488 (1.36, 8.94)**	**4.617 (2.53, 8.42)**
Trunk LM to total LM ratio (%)	0.846 (0.70, 1.02)	0.655 (0.44, 1.01)	0.737 (0.50, 1.09)	**0.675 (0.54, 0.84)**
Leg LM to total LM ratio (%)	1.048 (0.87, 1.26)	1.147 (0.79, 1.66)	1.187 (0.82, 1.72)	**1.310 (1.07, 1.61)**
%BF (%)	**0.891 (0.84, 0.95)**	**0.893 (0.80, 0.99)**	0.909 (0.82, 1.01)	**0.898 (0.85, 0.95)**
$\dot{V}O_{2peak}$ (ml·kg^{-1}·min^{-1})	**1.108 (1.04, 1.18)**	1.120 (0.99, 1.26)	1.108 (0.99, 1.24)	**1.132 (1.06, 1.21)**
Upper body power (W·kg^{-1})	**1.638 (1.23, 2.17)**	**3.910 (1.41, 10.81)**	**1.957 (1.01, 3.78)**	**2.061 (1.46, 2.91)**
Lower body power (W·kg^{-1})	**1.081 (1.02, 1.15)**	1.082 (0.95, 1.23)	1.062 (0.93, 1.21)	**1.106 (1.03, 1.19)**
1RM deadlift (kg)	**1.103 (1.06, 1.15)**	**1.148 (1.05, 1.26)**	**1.049 (1.00, 1.10)**	**1.104 (1.06, 1.16)**
1RM bench press (kg)	**1.204 (1.10, 1.32)**	**1.451 (1.12, 1.89)**	**1.086 (1.01, 1.17)**	**1.254 (1.12, 1.41)**
1RM back squat (kg)	**1.076 (1.04, 1.11)**	**1.076 (1.02, 1.13)**	1.031 (0.99, 1.07)	**1.069 (1.04, 1.10)**
Nonmodifiable				
Sex[a]	**0.007 (0.01, 0.06)**	-	0.137 (0.02, 1.21)	-
Age (y)	1.004 (0.92, 1.09)	**1.355 (1.04, 1.77)**	**1.270 (1.01, 1.61)**	1.004 (0.92, 1.09)
Height (cm)	**1.280 (1.17, 1.41)**	**1.142 (1.02, 1.28)**	**1.178 (1.04, 1.34)**	**1.211 (1.12, 1.31)**

Note: Significant predictors with confidence intervals are indicated in boldface.

Abbreviations: %BF, percent body fat; 1RM, one-repetition maximum; FM, fat mass; LM, lean mass; V̇O2peak, maximum volume of oxygen consumption in ml/kg/min.

[a]Sex was coded as 1 for women and 0 for men.

1RM deadlift (OR: 1.104 [1.06, 1.16]), %BF (OR: 0.898 [0.85, 0.95]), and 1RM back squat (1.069 [1.04, 1.10]) (Table 5).

3.2.5 | Maximum single lift

Significant characteristics predictive of passing MSL, from strongest to weakest, included upper body power (OR: 3.910 [1.41, 10.81]), trunk LM (OR: 2.387 [1.17, 4.88]), leg LM (OR: 2.290 [1.22, 4.31]), total LM (OR: 1.830 [1.15, 2.92]), 1RM bench press (OR: 1.451 [1.12, 1.89]), age (OR: 1.355 [1.04, 1.77]), 1RM deadlift (OR: 1.148 [1.12, 1.89]), body mass (OR: 1.146 [1.03, 1.28]), height (OR: 1.142 [1.02, 1.28]), %BF (OR: 0.893 [0.80, 0.99]), 1RM back squat (OR: 1.076 [1.02, 1.13]), and arm to total LM ratio (OR: 1.040 [1.01, 1.07]) (Table 5).

3.2.6 | Casualty drag

Significant characteristics predicting passing CD, from strongest to weakest, included arm LM (11.694 [3.84, 35.60]), arm to total LM ratio (OR: 3.734 [2.24, 6.23]), trunk LM (OR: 2.576 [1.70, 3.91]), leg LM (OR: 2.088 [1.57, 2.78]), male sex (OR: 0.007 [0.01, 0.06]), upper body power (OR: 1.638 [1.23, 2.17]), total LM (OR: 1.515 [1.27, 1.81]), height (OR: 1.280 [1.17, 1.41]), 1RM bench press (OR: 1.204 [1.10, 1.32]), leg FM (OR: 0.837 [0.72, 0.98]), body mass (OR: 1.142 [1.08, 1.21]), %BF (OR: 0.891 [0.84, 0.95]), $\dot{V}O_{2peak}$ (OR: 1.108 [1.04, 1.18]), 1RM deadlift (OR: 1.103 [1.06, 1.15]), lower body peak power (OR: 1.081 [1.02, 1.15]), and 1RM back squat (OR: 1.076 [1.04, 1.11]) (Table 5).

4 | DISCUSSION

This study aimed to identify the modifiable and nonmodifiable characteristics associated with passing GCC-related PES tasks and determine those task-specific modifiable characteristics fitted for overcoming nonmodifiable characteristics in a sample of men and women analogous to recruits eligible for GCC PES assessment. Our results revealed that relative and absolute arm LM, absolute trunk and leg LM, upper body power, and upper body maximal strength significantly predicted passing GCC-specific PES tasks involving dragging, carrying, throwing/pushing, and lifting heavy loads to meet our first hypothesis (H1). Additionally, it was observed that higher aerobic capacity, lower percent body fat, and greater lower body maximal strength were the strongest predictors for passing a material handling task, whereas higher aerobic capacity was found to significantly predict passing a load carriage task to contradict our second hypothesis (H2). Finally, it was observed that modifiable characteristics demonstrated higher predictive value than nonmodifiable characteristics in all PES tasks to support our final hypothesis (H3). Taken together, these findings may underscore the importance of modifiable characteristics and their role in potentially overcoming nonmodifiable limitations commonly observed in women to enhance GCC PES effectiveness.

The MBCT assesses a soldier's ability to perform upper body dominant tasks with rapid force production such as lifting fellow soldiers, throwing equipment over walls, and using force in hand-to-hand combat (Agostinelli et al., 2022). Height was found to be a significant nonmodifiable factor with a 21.1% increased likelihood of success for every 1-cm increase in value. However, several modifiable characteristics demonstrated stronger associations with task success including absolute and relative arm, trunk, and leg LM, upper body power, and upper body maximal strength (Table 5). For example, every 1 kg and 1% increase in absolute and relative arm LM was associated with a ~8.5-fold and ~4.6-fold increased likelihood of task success, respectively, while every 1 kg and 1 $W \bullet kg^{-1}$ increase in absolute leg LM and upper body power was associated with a ~2.5-fold and ~2-fold increased likelihood of task success. Previous research shows significant relationships between such modifiable characteristics on MBCT performance (Aandstad, 2020; Wilson et al., 2020). Wilson et al. (2020) observed a significant correlation ($r \geq 0.50$–0.69) between upper body LM and MBCT distance in male athletes (Wilson et al., 2020). Similarly, Aandstad et al. (Aandstad, 2020) found arm, trunk, and leg LM were significantly correlated ($r = 0.54$) with MBCT distance in military cadets (Aandstad, 2020). Upper body power assessed by explosive pushups (Harris et al., 2011) and bench press power throw (Cronin et al., 2004) were also associated with MBCT distance (Cronin et al., 2004; Harris et al., 2011). However, the association between MBCT distance and upper body maximal strength shows mixed results (Cronin et al., 2004; Ignjatovic et al., 2012). Nevertheless, previous research shows that targeting the sequential development of total LM followed by upper body maximal strength and power via periodized resistance or medicine ball throw training augments MBCT performance (Ignjatovic et al., 2012; Sterczala et al., 2023). Sterczala et al. (2023) observed a significant improvement in MBCT performance (+4.5%) after a 12-week periodized concurrent resistance and interval training program featuring four mesocycles of increased exercise intensity (64%–90% 1RM) to optimize rate of force development (Sterczala et al., 2023). These results, and results from similar studies (Ignjatovic et al., 2012), suggest the effectiveness of periodized resistance or medicine ball throw training to improve such modifiable characteristics for MBCT performance.

The MSL was utilized to assess the maximal mass lifted to a 1.49-m platform in which represents a practical lifting height akin to military truck load handling (Kraemer et al., 2001). Height was found to be significantly associated with task success as every 1-cm increase resulted in a 14.2% higher likelihood of a pass. Additionally, every 1-year increase in age was associated with approximately 36% increased likelihood of success (Table 5). However, specific modifiable characteristics, including upper body power, absolute trunk and leg LM, and upper body maximal strength, showed stronger predictive value. Upper body power showed the highest predictive value with every 1 $W \cdot kg^{-1}$ increase resulting in a 3.91-fold increased likelihood of passing this task. Similarly, every 1 kg increase in

absolute trunk and leg LM resulted in a 2.29 and 2.39-fold higher likelihood of success, respectively. Previous research supports the relationship between 1RM overhead lifting performance and upper body power (Soriano et al., 2022; Sterczala et al., 2023) and trunk (Zaras et al., 2020) and leg LM in weightlifting populations (Beck et al., 2019) as well as the effectiveness of targeting such characteristics to improve maximal lifting performance (Kraemer et al., 2001; Nindl et al., 2000, 2017; Sterczala et al., 2023). Nindl et al. (2000) observed significant improvements in total LM (+2.2%) and regional LM (leg LM: +5.5%; trunk LM: +1.0%; and arm LM: +0.6%) in civilian women after a periodized 24-week strength training program with an emphasis on overhead lifts (Nindl et al., 2000). This program also improved maximal lifting performance assessed by a 76.2–152.4 cm box lift (+47.0%) (Nindl et al., 2017). Notably, Kraemer et al. (2001) observed improvements in upper body strength, power, and 1RM box lift performance after a 4-month resistance training program involving ballistic, high-velocity movements in civilian women (Kraemer et al., 2001). Together, these results suggest the effectiveness of either periodized resistance training with or without ballistic movements to improve such modifiable characteristics for MSL outcomes and overcome stature limitations often observed in women (Beck et al., 2019; Drain et al., 2015; Kraemer et al., 2001; Nindl et al., 2017).

Casualty evacuation, a critical GCC task involving lifting, dragging, or carrying injured soldiers to safety, demands readiness from every soldier (Vaara et al., 2022). Two simulated tasks utilized in the current study, CD and WCC, required dragging and carrying a heavy load as quickly as possible. Simple binary logistic regression analysis demonstrated that male sex and height were significant nonmodifiable factors for CD, while both height and age favored success in WCC (Table 5). However, common modifiable characteristics, including absolute and relative arm and absolute trunk and leg LM, showed stronger predictive value than nonmodifiable factors in both tasks. These findings align with previous research on anthropometric characteristics associated with dragging and carrying performance (Beck et al., 2019; Lindberg et al., 2015). Total and regional LM, particularly leg LM, have been associated with carrying task performance in recruit populations (Beck et al., 2019). Hence, given that soldiers often carry body-borne loads exceeding 45 kg (Orr et al., 2021), a mass 20 kg greater than the load carried in the current study, targeting absolute and relative arm and absolute trunk and leg LM may augment casualty evacuation performance. Hendrickson et al. (2010) observed a significant relationship between changes in total LM and time to complete a 61.4 kg mannequin drag task ($r = -0.44$) in recreationally active women after a 12-week periodized strength training regimen with exercise selection targeting large muscle groups (Hendrickson et al., 2010). Sterczala et al. (2023) observed a 29.8% and 13.8% improvement in CD and WCC performance, respectively, after a 12-week periodized concurrent resistance training regimen involving task-specific loaded carries (Sterczala et al., 2023). These results may suggest the effectiveness of either total body or task-specific resistance training to improve casualty evacuation performance.

Manual material handling is a common task performed by soldiers in addition to load carriage and casualty evacuation (Vaara et al., 2022). Backward stepwise multiple logistic regression analysis revealed that higher aerobic capacity and lower body maximal strength and lower body fat percentage significantly predicted RLC success independent of sex (Table 3). Repeated lifting and carrying task performance continuous and repetitive in nature has shown to benefit from sufficient aerobic capacity for reduced peripheral fatigue and musculoskeletal injury risk (Savage et al., 2012). The RLC in the current study comprised a time limit of 20 min, which may have allowed fitter participants to maintain a high work output. Previous reports show that greater lower body maximal strength enhances military occupational repeated lift and carry task performance (Kraemer et al., 2001), which can be supported by enhanced directional change efficiency (Hammami et al., 2018). Finally, lower percent body fat has been associated with improved loaded military simulation tests involving repeated lifts (Vaara et al., 2022). Enhancement of aerobic capacity and lower body maximal strength may be accomplished via concurrent resistance and low-volume, high-intensity interval training (Burley et al., 2020; Kraemer et al., 2001), which may augment manual material handling performance (Burley et al., 2020; Sterczala et al., 2023).

Aerobic capacity emerged as the strongest predictor for passing the 2KRM to contrast our second hypothesis (Table 4). This finding contradicts previous studies highlighting the importance of both upper and lower body strength and aerobic capacity for efficient load carriage performance (Fallowfield et al., 2012; Knapik et al., 2004). A systematic review showed that the combination of total body strength and aerobic training with load carriage yielded significant improvement in load carriage performance (Knapik et al., 2012a). Fallowfield et al. (2012) observed that load carriage performance required both aerobic fitness and upper and lower body muscular strength when carrying heavier loads up to 31.0 kg (Fallowfield et al., 2012). However, unlike previous work (Fallowfield et al., 2012; Knapik et al., 2012b), the load carried in the current study (25 kg) proved lighter than previous loads (31–46 kg). Moreover, the distance covered during the 2KRM exceeded previous distances of 100–800 m and 0.4 km during similar tasks (Harman et al., 2008; Knapik et al., 2012b). Thus, the combination of increased load carriage distance with lighter carrying loads used in the current study may infer the utilization of oxidative means to achieve 2KRM success (Burley et al., 2020). Enhancements in aerobic capacity may be accomplished via concurrent resistance and low-volume, high-intensity interval training (Burley et al., 2020), or aerobic training (Kraemer et al., 2001) to accommodate increases in LM.

In terms of strengths, our study serves as a valuable extension of prior research (Ahamed et al., 2021; Conkright et al., 2022b; Sterczala et al., 2023), contributing significant insights into the physical attributes required for successfully completing GCC-based tasks in women (Epstein et al., 2013). Furthermore, our sample was carefully selected to represent incoming recruits based on demographics and physical activity levels (Hall et al., 2022). Although 1RM and relative power values in the current sample were lower than athletic

populations, these results highlight the feasibility of increasing maximal strength for PES task performance while maintaining or developing appropriate aerobic capacity utilizing appropriate training methods. Additionally, all GCC-based PES tasks utilized in the current study represented tasks based on and adapted from protocols from those recently developed by the British Army for GCC roles now implemented as RFT(S), RFT(E), and RFT(BT). Although the protocols used in the current study were similar to RFT(S), RFT(E), and RFT(BT) and did not reflect the final protocols and standards adopted by the British Army, the employment of such tasks for the PES assessment attained ecological relevance to reflect military occupational tasks frequently performed in GCC roles (Fitriani et al., 2016).

However, it is essential to acknowledge the limitations of the current study. Notably, we did not monitor exercise intensity during the PES assessment and physical testing battery although participants were explicitly instructed to exert their maximum effort. Additionally, the inclusion of a 48-h recovery period between testing sessions may have had an impact on peripheral fatigue and performance. However, previous research has shown that a 48-h recovery period can help maintain maximal exercise performance in active populations (Timón et al., 2019). Further, the height of the box used for the weighted squat jump protocol was visually adjusted for each participant to achieve a 90˚ bend at the knee without using an objective method to determine the starting knee angle (e.g., goniometry). Therefore, there may have been some participants in the starting squat position with a few degrees lesser or greater than 90˚, which should not significantly alter the outcome. Additionally, the hip and ankle joint angle performed at the starting squat position was not controlled but variable dependent upon the participant where some participants may have started the squat with a more upright posture with greater dorsiflexion and others more flexed at the hip joint. However, this enabled participants to utilize their optimal mechanics given their system (e.g., body) constraints. Moreover, all participants were civilians without prior military training experience or completion before enrolling in the current study. The completion of military training could affect PES performance as the tests utilized reflect military occupational tasks that recruits or trained soldiers would be trained to perform. Therefore, the observed PES performance may not be generalizable to individuals with prior military training experience or completion. Nevertheless, our results reveal pertinent modifiable characteristics that remain relevant to target for improvement in both civilian women and seasoned female soldiers in which may benefit GCC-based PES performance regardless of military experience.

5 | CONCLUSION

Passing a GCC-based PES assessment involving tasks requiring lifting, dragging, pushing, throwing, and carrying heavy loads may require sufficient upper body (arm and trunk) and lower body LM, upper body power, and upper body maximal strength in recruits eligible for GCC roles. Passing a GCC-based manual material handling task may be associated with reduced body fat percentage and enhanced lower body muscular strength and aerobic capacity with the latter important for passing a load carriage task. Targeting modifiable characteristics pertinent for pushing/throwing and maximal lifting tasks assessed by MBCT and MSL, respectively, including upper and lower body LM, strength, and power, may be improved via periodized resistance training for the sequential development of LM followed by strength and power for the rate of force development. Targeting modifiable characteristics pertinent for carrying heavy loads assessed by CD and WCC, including relative and absolute arm LM and absolute trunk and leg LM, may be improved via total body or task-specific resistance training. Concurrent resistance and high-intensity interval training or aerobic training may develop aerobic capacity to accommodate LM for manual material handling and load carriage tasks assessed by RLC and 2KRM, respectively. Modifiable characteristics may hold greater predictive value than nonmodifiable delimiters, including age, height, and sex, on passing GCC PES tasks, indicating an opportunity to target these characteristics for improving GCC-based PES performance in women.

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CONFLICT OF INTEREST STATEMENT

No conflicts of interest, financial or otherwise, are declared by the authors. The results of the current investigation do not constitute endorsement of the product by the authors, the U.K. Ministry of Defense, U.S. Department of Defense, or the U.S. Government. Citations of any commercial organizations or trade names mentioned in the current investigation do not constitute an official endorsement or approval of the products or services of those organizations.

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